

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Tech. Examination (EC)

November 2013

EE221 ELECTRICAL TECHNOLOGY

Date: 30.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain following parts of DC machine with diagram (1) yoke (2) pole core and pole shoe (3) armature core and armature winding. [07]

Q - 2 (a) What is the necessary of starter in DC machine? Explain three point and four point starters with circuit diagram. [08]

(b) Derive the equivalent circuit of a transformer. [06]

OR

Q - 2 (a) What is all day efficiency of transformer? Derive condition for maximum efficiency of transformer. [07]

(b) A 4kva 200v/400v transformer supply the load current at power factor of 0.8 lagging. The test results are OC test: 200v, 0.8A, 70W LV side and SC Test: 20 v, 10A, 60W HV side. Calculate efficiency and voltage regulation of transformer at half load and 0.8 p.f. lagging. [07]

Q - 3 (a) Explain shaded pole induction motor with necessary diagrams. [08]

(b) Define power factor and explain importance of power factor improvement for consumers and generating station. [06]

OR

Q - 3 (a) Explain single value and double value capacitor start and run induction motor with circuit diagram. [08]

(b) Give the names of equipments to be used for power factor improvement. Explain any one of them. [06]

SECTION - II

Q - 4 (a) Answer the following questions. [07]

(1) Give the names of insulators used in substation.

(2) What is the function of switching substation?

- (3) Define synchronous impedance of alternator.
- (4) Define voltage regulation of alternator.
- (5) Give the equation of speed of rotor field of induction motor.
- (6) State the names of methods of speed control of induction motor from stator side.
- (7) What is the condition for maximum torque of induction motor?

- Q - 5 (a)** Explain double cage induction motor with rotor construction. [07]
- (b)** Explain auto transformer starter for induction motor with circuit diagram. [07]

OR

- Q - 5 (a)** Explain rotor rheostat method and cascade connection method for speed control of induction motor. [07]
- (b)** A three phase star connected alternator is rated at 1200kva, 12000v the armature resistance and synchronous reactance are 5Ω and 35Ω respectively. Calculate regulation of alternator by using phase diagram for a load of 960kw at 0.8 p.f. lagging. [07]

- Q - 6 (a)** Draw stator and rotor construction of alternator and explain the same. [07]
- (b)** Explain transformer substation with flow of power through different voltage levels. [07]

OR

- Q - 6 (a)** Explain laboratory methods to calculate armature resistance (R_a) and synchronous impedance (Z_s) of alternator. [07]
- (b)** Draw the layout of pole mounted s/s and explain function of each equipment. [07]
